B V N S HEMANTH

Founding Engineer — AI

bharatulahemanth@gmail.com · +91 99126 39497 · linkedin.com/in/bharatula-hemanth-6a3782236 · github.com/hementhu · hementhu.github.io · India · remote, open to relocation

Founding engineer who took an LLM decision-intelligence product from an empty repo to production in seven months: wrote 80–90% of the calculation and inference core, the runtime validators that check every generated claim against source data, and the mutation-tested evaluation proving those checks work. Owns roadmap, scoping and client meetings alongside the code. IIT Madras. Won a 15-team Physical AI hackathon; 1st place, WISER 2026 BQP challenge.

EXPERIENCE

infigro.ai · Founding Engineer (first hire) — AI & Data — Feb 2026 – Present

Decision-intelligence platform for growth leaders at OTT and e-commerce companies: analyses subscriber and revenue data end to end, then generates insights and prescribed actions with LLMs. Team of 5.

- Built **80–90% of the core calculation and inference layer** the rest of the team builds on: ingestion, sanitisation, EDA, decomposition and mix-shift analysis, chart and report rendering across **70+ analysis types**.
- Shipped **v1 to production in Sep 2026**, seven months from an empty repo. In daily use at a national OTT platform (paid contract from Oct 2026); two further OTT platforms committed, contracts in signing.
- Designed **runtime grounding validators** that check every number, filter and claim in LLM-generated output against the source data before delivery. Fail-closed: a failing output is retried or rejected, never shown.
- Proved the validators with **mutation testing: 52 of 52 seeded defects caught, 0 false positives**; golden regression suites gate every release.
- Cut report generation from **40 to 12 minutes** on **10M+ row** client datasets (data layer CSV/pandas → Parquet).
- Beyond code: wrote the **product roadmap**, scoped features with the founders, run **client meetings** with the live customer's growth team, and mentor the junior engineer.
- Designed the **prompt architecture**: typed output contracts with deterministic repair paths, so malformed LLM outputs self-correct at runtime instead of failing client runs.

Jaguar Land Rover · Graduate Engineer, R&D — Jul 2024 – Feb 2026 (notice period to Apr 2026)

- Built an **LLM pipeline (Flan-T5-XXL)** that deduplicated 3,000 interview questions, replacing ~4.5M pairwise manual comparisons. Random-sample audits found no errors; ran weekly through hiring cycles with zero production failures. **100+ hours saved per run.**
- Replaced FEM/CFD simulations with **physics-informed neural network surrogates**: **18× faster inference** (5 s vs 90 s) at ~1e-5 error. 4th of 40 in the internal R&D showcase; awarded AWS credits and conference sponsorship.
- Modernised **100+ legacy C files** and replaced VBA tooling with a Python DCM parser (70% less processing time); integrated into the Roll Stability Control pipeline for 2027 vehicles.

AWARDS & PATENTS

- **Winner (1st of 15), B1's Hacktober: Physical AI 2026** — led one-shot imitation on an SO-101 arm (one human video → validated action plan, no teleop data or training); task time ~7 → 2–3 min vs Claude Code driving the arm. Prize: LeRobot arm + Bengaluru Tech Summit 2026 pitch.
- **1st place (solo), WISER 2026 × BQP challenge** — showed a claimed quantum-PINN advantage is classically reproducible (111 models, fair controls); invited to the fellowship interview. **arXiv preprint (quant-ph, 2026).**
- **Co-inventor, UK patent filed 2024** — Simulated Driving Experience using Integrated Vehicle Systems; a second is in filing.

OPEN SOURCE & WRITING

- **graphify** (126k★, codebase → knowledge graph) — landed a v0.9.92 fix for Python node-ID collisions that silently dropped symbols (e.g. `requests.Session` vs `session()`).
- **codemap** — open-source visual map of any Python codebase from pure AST analysis, no LLM; every claim links to its exact source lines. github.com/hementhu/codemap
- **Writing on LLM agent reliability** (LinkedIn) — *Unapologetic Sorry*; *The model explains the decision, it doesn't make it*.

SKILLS

Build	Python, SQL, PyTorch, FastAPI, AWS, Parquet/pandas, ECharts/Plotly, Git, C; OpenCV, MediaPipe
LLM systems	agent orchestration, typed output contracts, runtime validation, mutation-tested evaluation, prompt architecture
Analysis	decomposition and mix-shift analysis, causal-claim guards, EDA at 10M+ rows, statistics and optimisation

EDUCATION

B.Tech, Mechanical Engineering, Minor in AI/ML — Indian Institute of Technology Madras, 2024 · CGPA 8.25/10.